

SSH Brute Force Detection and Automated Response Using Wazuh

Security Operations Center (SOC) Incident Detection & Response Project

Executive Summary

This project demonstrates the implementation of a Security Information and Event Management (SIEM) use case using Wazuh to detect, investigate, and automatically respond to SSH brute-force attacks targeting a Linux endpoint.

The lab environment consists of a Kali Linux attacker machine, an Ubuntu Linux endpoint configured with the Wazuh Agent, and a Wazuh Manager deployed using Docker. Multiple failed SSH authentication attempts were simulated from the attacker machine to replicate a brute-force attack scenario. Wazuh successfully detected the suspicious activity, correlated authentication failures, generated security alerts, mapped the attack to the MITRE ATT&CK framework, and executed an automated Active Response action to block the attacker's IP address.

This project demonstrates practical skills in security monitoring, log analysis, detection engineering, incident investigation, threat classification, and automated incident response.

Project Objectives

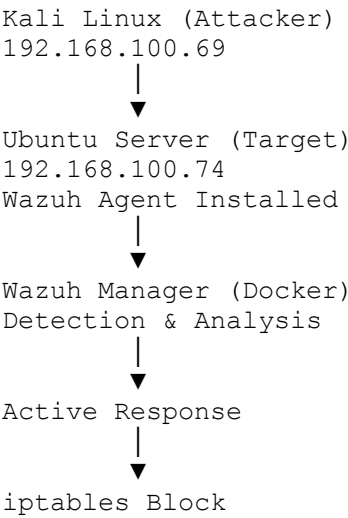
The primary objectives of this project were:

- Monitor SSH authentication activity on a Linux endpoint.
 - Detect brute-force authentication attempts.
 - Develop custom Wazuh detection rules.
 - Implement MITRE ATT&CK mapping.
 - Configure Active Response for automated containment.
 - Investigate security events from detection to response.
 - Simulate a real-world SOC workflow.
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Lab Environment

Component	Purpose
Kali Linux	Attack Simulation
Ubuntu Linux	Target Endpoint
Wazuh Agent	Log Collection
Wazuh Manager (Docker)	Event Analysis & Correlation
SSH Service	Authentication Target
iptables	Automated Containment

Network Architecture



Phase 1: Wazuh Agent Deployment

The Ubuntu endpoint was configured with the Wazuh Agent to collect and forward security logs to the Wazuh Manager.

The agent establishes secure communication with the manager and enables monitoring of authentication events occurring on the endpoint.

Phase 2: Attack Simulation

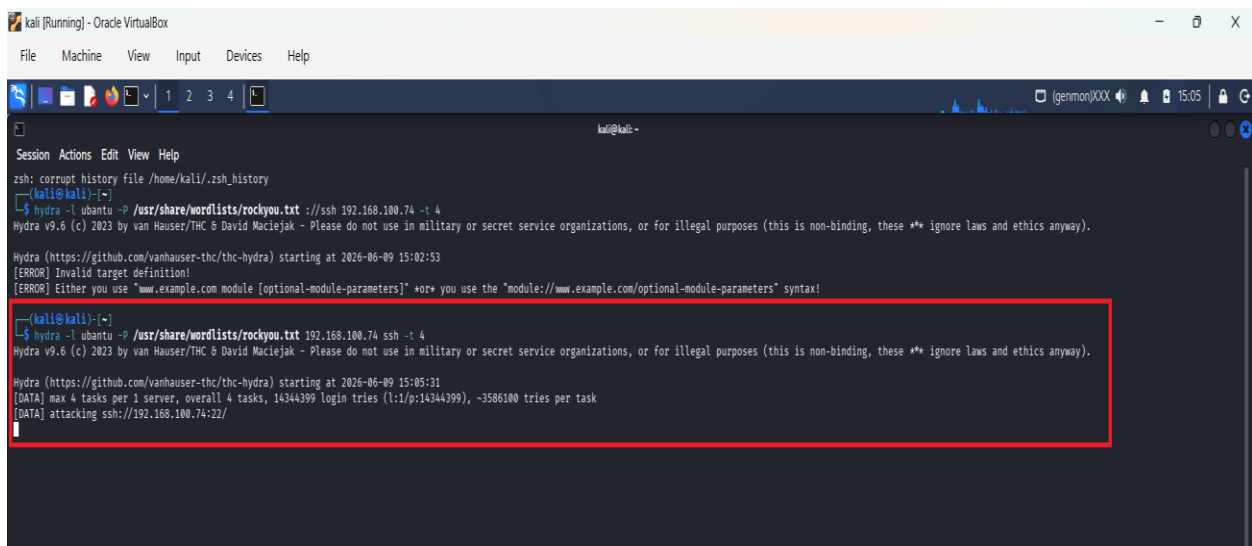
A brute-force attack simulation was performed from the Kali Linux machine by generating multiple failed SSH authentication attempts against the Ubuntu server.

The objective was to create authentication failures that could be detected and analyzed by Wazuh.

Attack Characteristics

- Multiple failed login attempts
- Repeated authentication failures
- Same source IP address
- Short time interval between attempts

Screenshot 1 – SSH Authentication Attempts

A screenshot of a Kali Linux terminal window. The terminal shows the execution of the Hydra tool to perform a brute-force attack on an SSH server. The command used is 'hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt -t 4 ssh 192.168.100.74'. The output shows the Hydra tool starting at 2026-06-09 15:02:53, followed by an error message 'Invalid target definition!'. A second command is then executed: 'hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt 192.168.100.74 ssh -t 4'. The output for this command shows Hydra starting at 2026-06-09 15:05:31, with data indicating 4 tasks, 14344399 login tries, and 3586100 tries per task. The terminal window has a dark background and a red border around the command and output text.

```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt -t 4 ssh 192.168.100.74
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-09 15:02:53
[ERROR] Invalid target definition!
[ERROR] Either you use "www.example.com module [optional-module-parameters]" *or* you use the "module://www.example.com/optional-module-parameters" syntax!

kali@kali:~$ hydra -l ubuntu -P /usr/share/wordlists/rockyou.txt 192.168.100.74 ssh -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-09 15:05:31
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (1:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.100.74:22/
```

01_Attack_Simulation.png

Purpose:

Shows SSH authentication attempts originating from the attacker machine.

Phase 3: Log Collection and Analysis

The Ubuntu endpoint recorded authentication events in:

/var/log/auth.log

These logs were collected by the Wazuh Agent and forwarded to the Wazuh Manager for analysis.

Example events included:

- Failed password attempts
- Invalid user authentication
- Repeated login failures

Phase 4: Detection by Wazuh

Wazuh analyzed the authentication events and identified repeated failed login attempts.

A built-in SSH authentication detection rule generated alerts indicating suspicious authentication behavior.

Detected Event

Rule ID: 2502

Description:

User missed the password more than one time

Severity Level: 10

MITRE ATT&CK Technique:

T1110 - Brute Force

Screenshot 2 – Wazuh Detection Alert

1,796 hits					
Jun 8, 2026 @ 14:38:25.953 - Jun 9, 2026 @ 14:38:25.953					
Export Formatted Reset view 759 available fields Columns Density 1 fields sorted Full screen					
timestamp	agent.name	rule.description	rule.level	rule.id	
Jun 9, 2026 @ 14:36:10.763	ubuntu	syslog: User missed the password more than one time	10	2502	
Jun 9, 2026 @ 14:36:10.756	ubuntu	syslog: User missed the password more than one time	10	2502	
Jun 9, 2026 @ 14:36:10.747	ubuntu	syslog: User missed the password more than one time	10	2502	
Jun 9, 2026 @ 14:36:10.739	ubuntu	syslog: User missed the password more than one time	10	2502	
Jun 9, 2026 @ 14:36:08.771	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:08.765	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:08.764	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:08.746	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:08.737	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:06.734	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:06.733	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:06.731	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:02.738	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:02.736	ubuntu	sshd: authentication failed.	5	5760	
Jun 9, 2026 @ 14:36:02.732	ubuntu	sshd: authentication failed.	5	5760	

02_Wazuh_Detection.png

Context of #Va29q548Twpzc7SU-ps	
> Jun 9, 2026 @ 14:36:10.763	predecoder.hostname: ubuntu predecoder.program_name: sshd predecoder.timestamp: Jun 09 09:36:10 input.type: log agent.ip: 192.168.100.74 agent.name: ubuntu agent.id: 002 manager.name: wazuh.manager data.srcip: 192.168.100.69 data.dstuser: ubuntu rule.mail: false rule.level: 10 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 16 4.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User missed the password more than one time rule.groups: syslog, access_control, authentication_fa iled rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.35.7.d, IV.32.2 rule.firedtimes: 7 rule.mitre.technique: Brute Force rule.mitre.id: T1110 rule.mitre.tactic: Credentia l Access rule.id: 2502 rule.pgpi3: 7.8 location: journalld decoder.parent: sshd decoder.name: sshd id: 1780997770.3445925 full_log: Jun 09 09:36:10 ubuntu sshd[6972]:
> Jun 9, 2026 @ 14:36:10.756	predecoder.hostname: ubuntu predecoder.program_name: sshd predecoder.timestamp: Jun 09 09:36:10 input.type: log agent.ip: 192.168.100.74 agent.name: ubuntu agent.id: 002 manager.name: wazuh.manager data.srcip: 192.168.100.69 data.dstuser: ubuntu rule.mail: false rule.level: 10 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 16 4.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User missed the password more than one time rule.groups: syslog, access_control, authentication_fa iled rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.35.7.d, IV.32.2 rule.firedtimes: 6 rule.mitre.technique: Brute Force rule.mitre.id: T1110 rule.mitre.tactic: Credentia l Access rule.id: 2502 rule.pgpi3: 7.8 location: journalld decoder.parent: sshd decoder.name: sshd id: 1780997770.3445388 full_log: Jun 09 09:36:10 ubuntu sshd[6973]:
> Jun 9, 2026 @ 14:36:10.747	predecoder.hostname: ubuntu predecoder.program_name: sshd predecoder.timestamp: Jun 09 09:36:10 input.type: log agent.ip: 192.168.100.74 agent.name: ubuntu agent.id: 002 manager.name: wazuh.manager data.srcip: 192.168.100.69 data.dstuser: ubuntu rule.mail: false rule.level: 10 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 16 4.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User missed the password more than one time rule.groups: syslog, access_control, authentication_fa iled rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.35.7.d, IV.32.2 rule.firedtimes: 5 rule.mitre.technique: Brute Force rule.mitre.id: T1110 rule.mitre.tactic: Credentia l Access rule.id: 2502 rule.pgpi3: 7.8 location: journalld decoder.parent: sshd decoder.name: sshd id: 1780997770.3444851 full_log: Jun 09 09:36:10 ubuntu sshd[6971]:
> Jun 9, 2026 @ 14:36:10.739	predecoder.hostname: ubuntu predecoder.program_name: sshd predecoder.timestamp: Jun 09 09:36:10 input.type: log agent.ip: 192.168.100.74 agent.name: ubuntu agent.id: 002 manager.name: wazuh.manager data.srcip: 192.168.100.69 data.dstuser: ubuntu rule.mail: false rule.level: 10 rule.pci_dss: 10.2.4, 10.2.5 rule.hipaa: 16 4.312.b rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: syslog: User missed the password more than one time rule.groups: syslog, access_control, authentication_fa iled rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.35.7.d, IV.32.2 rule.firedtimes: 4 rule.mitre.technique: Brute Force rule.mitre.id: T1110 rule.mitre.tactic: Credentia l Access rule.id: 2502 rule.pgpi3: 7.8 location: journalld decoder.parent: sshd decoder.name: sshd id: 1780997770.3444314 full_log: Jun 09 09:36:10 ubuntu sshd[6970]:
> Jun 9, 2026 @ 14:36:08.77	predecoder.hostname: ubuntu predecoder.program_name: sshd predecoder.timestamp: Jun 09 09:36:08 input.type: log agent.ip: 192.168.100.74 agent.name: ubuntu agent.id: 002 manager.name: wazuh.manager data.srcip: 192.168.100.69 data.dstuser: ubuntu data.srcport: 60910 rule.mail: false rule.level: 5 rule.hipaa: 164.312.b rule.pci_dss: 10.2.4, 10.2.5 rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication failed. rule.groups: syslog, sshd, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV.35.7.d, IV.32.2 rule.firedtimes: 5 rule.mitre.technique: Password Guessing, SSH rule.mitre.id: T1110.001, T1021.004

03_Wazuh_Detail_Detection.png

Purpose:
Shows the initial brute-force detection alert generated by Wazuh.

Phase 5: Detection Engineering

To improve detection accuracy and reduce false positives, a custom Wazuh correlation rule was created.

File Modified:

/var/ossec/etc/rules/local_rules.xml

Custom Rule:

```
<rule id="100500" level="12" frequency="10" timeframe="120">
  <if_matched_sid>2502</if_matched_sid>
  <description>Potential SSH Brute Force Attack Detected</description>

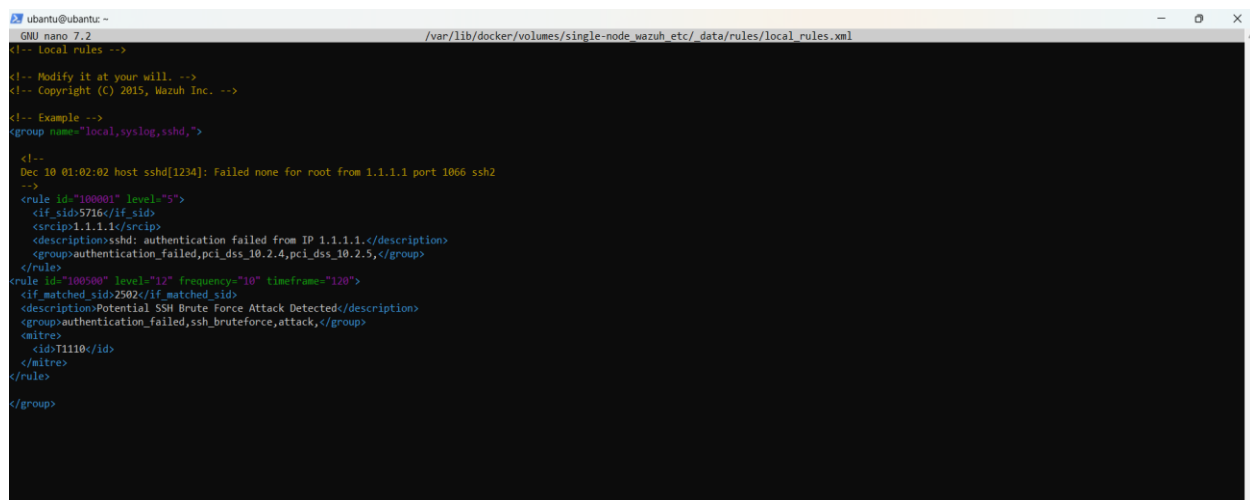
  <group>authentication_failed,ssh_bruteforce,attack,</group>

  <mitre>
    <id>T1110</id>
  </mitre>
</rule>
```

Detection Logic

- Monitor Rule 2502 events.
- Trigger after 10 events.
- Events must occur within 120 seconds.
- Increase alert severity.
- Apply MITRE ATT&CK classification.

Screenshot 4 – Custom Rule Configuration



Purpose:

Demonstrates creation of a custom brute-force detection rule.

Phase 6: MITRE ATT&CK Mapping

The custom rule was mapped to the MITRE ATT&CK framework.

Tactic	Technique	ID
Credential Access	Brute Force	T1110

MITRE mapping provides standardized threat classification and improves analyst understanding of attacker behavior.

Phase 7: Active Response Configuration

To automatically contain detected attacks, Wazuh Active Response was configured.

File Modified:

```
/var/ossec/etc/ossec.conf
```

Configuration:

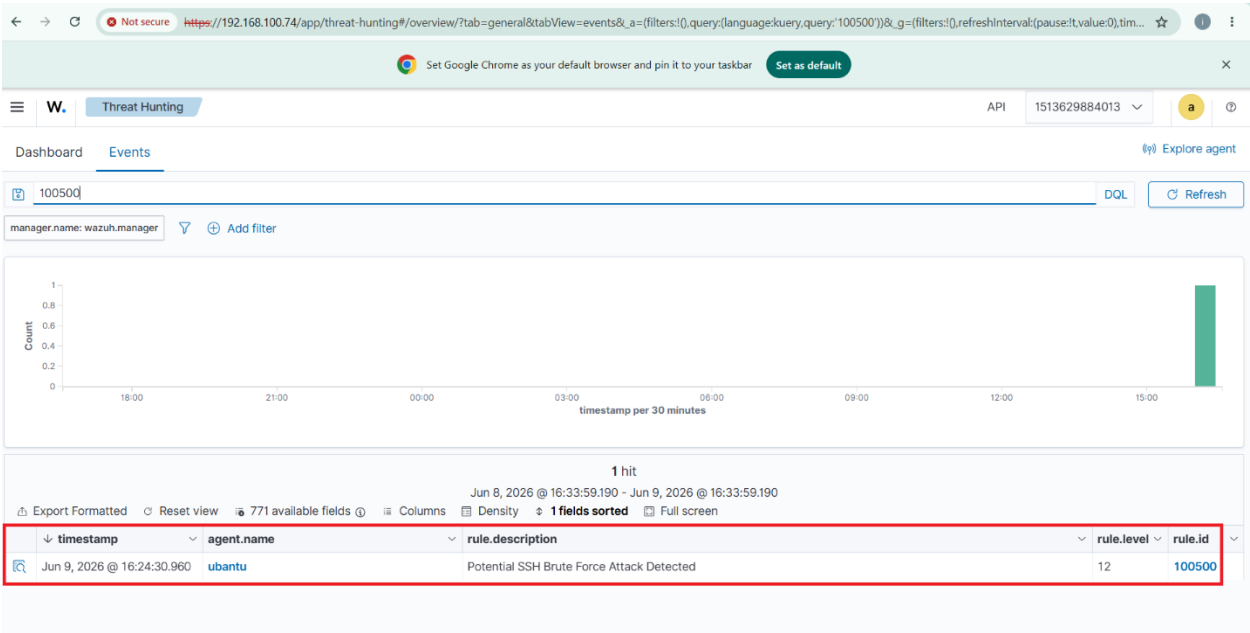
```
<active-response>
  <disabled>no</disabled>
  <command>firewall-drop</command>
  <location>local</location>
  <rules_id>100500</rules_id>
  <timeout>300</timeout>
</active-response>
```

Response Logic

- Trigger on Rule 100500.
- Execute firewall-drop script.
- Block attacker IP address.
- Maintain block for 300 seconds.

Screenshot 5 – Active Response Configuration

Screenshot 6 – Active Response Log



06_Active_Response_Log.png

Purpose:
Shows successful execution of Active Response actions.

Phase 9: Containment Verification

Firewall rules were reviewed to verify successful containment.

Command Used:

```
sudo iptables -L -n
```

Verification confirmed that the attacker's IP address had been blocked.

Incident Timeline

Time Event

- 09:35 Brute-force activity initiated
- 09:35 Authentication failures recorded

Time Event

09:36 Rule 2502 triggered

09:36 Rule 100500 triggered

09:36 Active Response executed

09:36 Attacker IP blocked

Key Findings

- Wazuh successfully detected repeated SSH authentication failures.
 - Custom detection logic improved detection accuracy.
 - MITRE ATT&CK mapping enhanced threat visibility.
 - Active Response enabled automated containment.
 - Firewall rules successfully blocked the attack source.
 - The project demonstrated a complete detection-to-response workflow.
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Skills Demonstrated

- SIEM Administration
 - Wazuh Configuration
 - Linux Log Analysis
 - Detection Engineering
 - Alert Correlation
 - MITRE ATT&CK Mapping
 - Threat Detection
 - Security Monitoring
 - Incident Investigation
 - Incident Response
 - Active Response Automation
 - Firewall Management
 - Security Operations Center (SOC) Workflow
-

Learning Outcomes

Through this project, the following technical and analytical skills were developed:

- Understanding of SSH authentication attack patterns.
- Practical experience with Wazuh SIEM deployment and management.

- Creation of custom correlation rules for advanced threat detection.
 - Implementation of MITRE ATT&CK-based threat classification.
 - Investigation of security events using centralized logging.
 - Configuration and validation of automated incident response mechanisms.
 - Application of firewall-based containment strategies.
 - Experience following a real-world SOC detection, investigation, and response workflow.
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Conclusion

This project successfully demonstrated the detection, analysis, and automated containment of SSH brute-force attacks using Wazuh. By integrating custom detection engineering, MITRE ATT&CK mapping, and Active Response automation, the solution provides an effective framework for identifying and mitigating unauthorized access attempts.

The project reflects practical SOC analyst responsibilities and showcases core cybersecurity skills including monitoring, investigation, detection engineering, threat classification, and incident response.